

SIMATS ENGINEERING

ASSESSMENT TOOL 2

**COURSE CODE: ITA 06
LEARNING**

COURSE NAME: MACHINE

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

*I SARSHINI .S 192321113 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

SARSHINI

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process.

1. Functional Blocks of a Machine Learning System

A typical Machine Learning (ML) system consists of the following key functional blocks:

a) Data Collection

- The first step involves gathering raw data from various sources such as databases, sensors, websites, or user inputs.
- Example: Collecting student marks, customer purchase data, or medical records.

b) Data Preprocessing

- Raw data is cleaned and transformed to remove noise, missing values, and inconsistencies.
- Includes normalization, handling missing data, and encoding categorical variables.
- Example: Filling missing marks with averages or removing duplicate records.

c) Feature Extraction (Feature Engineering)

- Important attributes (features) are selected or created from raw data to improve model performance.
- Reduces dimensionality and focuses on relevant information.
- Example: From student data, extracting features like average marks, attendance percentage.

d) Model Training

- The processed data is used to train a machine learning model using algorithms such as Decision Trees, SVM, or Neural Networks.
- The model learns patterns and relationships from the data.

e) Model Evaluation

- The trained model is tested using new (unseen) data to evaluate its performance.
- Metrics such as accuracy, precision, recall, or clustering quality are used.
- Ensures the model generalizes well and avoids overfitting.

2. Working of Machine Learning System

The workflow can be summarized as:

Data Collection → Preprocessing → Feature Extraction → Model Training → Evaluation → Prediction

3. Supervised Learning Example

- **Definition:** Uses **labelled data** (input + correct output).
- **Example:** Predicting whether a student will pass or fail.

Input (Marks, Attendance) Output (Label)

80, 90% Pass

40, 50% Fail

- The model learns the mapping between input features and known outputs.
- During testing, it predicts the output for new student data.

Key Point:

✓ Data is labelled

✓ Goal is prediction/classification

4. Unsupervised Learning Example

- **Definition:** Uses **unlabelled data** (no predefined output).
- **Example:** Customer segmentation.

Customer Data (Age, Income, Spending)

25, 30K, High

45, 80K, Medium

- The model groups similar data into clusters using algorithms like K-Means.
- No predefined labels are provided.

Key Point:

- ✓ Data is unlabelled
- ✓ Goal is pattern discovery or clustering

Conclusion

A Machine Learning system transforms raw data into meaningful predictions or patterns through structured stages such as preprocessing, feature extraction, training, and evaluation. Supervised learning relies on labelled data for prediction tasks, while unsupervised learning works with unlabelled data to uncover hidden structures and relationships.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response.

1. Introduction

A Machine Learning (ML) system for disaster management aims to predict natural disasters such as floods and earthquakes using both historical and real-time data. Such a system helps in minimizing damage, saving lives, and enabling timely decision-making.

2. System Architecture

Workflow:

Data Collection → Preprocessing → Feature Selection → Model Training → Prediction → Alert System

3. Data Sources

a) Historical Data

- Past records of floods, earthquakes, rainfall, seismic activity
- Government databases, meteorological departments

b) Real-Time Data

- Sensor data (water level sensors, seismic sensors)
- Satellite imagery and weather APIs
- IoT devices and remote sensing systems

4. Data Preprocessing Steps

- **Data Cleaning:** Remove missing, inconsistent, or noisy data
- **Normalization:** Scale features (e.g., rainfall, magnitude) to a standard range
- **Time-Series Alignment:** Synchronize data collected at different intervals
- **Handling Missing Values:** Use interpolation or mean substitution
- **Data Integration:** Combine multiple sources (sensor + satellite + historical)

5. Feature Selection Techniques

- **Correlation Analysis:** Identify features strongly related to disaster occurrence
- **Principal Component Analysis (PCA):** Reduce dimensionality
- **Feature Importance (Tree-based models):** Select impactful variables
- **Domain Knowledge:** Example features:
 - Flood: rainfall intensity, river level, soil moisture
 - Earthquake: seismic wave frequency, magnitude, fault line proximity

6. Model Selection and Justification

a) Neural Networks (Recommended)

- **LSTM (Long Short-Term Memory):**
 - Suitable for time-series data (rainfall patterns, seismic signals)
 - Captures temporal dependencies
- **CNN (Convolutional Neural Networks):**
 - Useful for satellite image analysis (flood mapping)

b) Other Algorithms

- **Random Forest:**
 - Handles non-linear relationships
 - Good for structured/tabular data
- **Support Vector Machine (SVM):**
 - Effective for classification (disaster vs no disaster)
- **K-Means:**
 - Clustering high-risk zones

Justification:

Neural networks (especially LSTM) are ideal due to their ability to process sequential and real-time data, making them highly suitable for disaster prediction.

7. Training, Validation, and Evaluation

a) Training

- Split dataset into training data
- Train model using historical data patterns

b) Validation

- Use validation dataset to tune hyperparameters
- Prevent overfitting

c) Testing/Evaluation

- Evaluate model using unseen data
- Metrics:
 - Accuracy
 - Precision & Recall
 - F1-Score
 - RMSE (for prediction errors)

8. Deployment and Prediction

- Integrate model with real-time data streams
- Continuously update predictions
- Trigger alerts when risk threshold is exceeded

9. Conclusion

A machine learning-based disaster management system leverages historical and real-time data to provide accurate predictions and timely alerts. Using advanced models like LSTM and CNN improves performance in handling time-series and spatial data, enabling effective early warning, risk mitigation, and disaster response.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand